



H1 BIOLOGY Paper 1 and 2 **ANSWER SCHEME**

Paper 1

1	B	11	D	21	C
2	B	12	D	22	B
3	A	13	B	23	D
4	A	14	D	24	D
5	B	15	B	25	C
6	C	16	D	26	A
7	B	17	C	27	D
8	B	18	C	28	C
9	B	19	D	29	D
10	B	20	B	30	B

Paper 2

- Fig. 1.1 shows the tertiary structure of a molecule Ribonuclease, an enzyme that degrades RNA. The letter C represents cysteine, a sulphur containing amino acid. The numbers represent positions along the polypeptide chain. Number 40, for example, is the 40th amino acid.

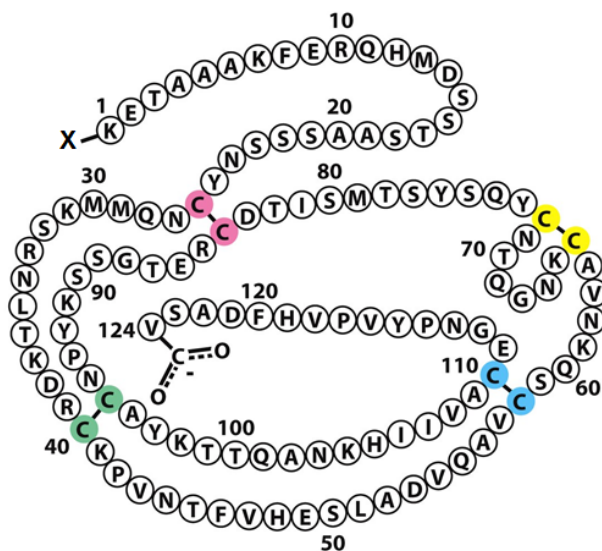


Fig. 1.1

(a) With reference to Fig. 1.1,

- Name the chemical group represented by the letter X, attached to the first amino acid. [1]

- $\text{NH}_2 / \text{NH}_3^+$ / amine group

- Explain why all molecules of Ribonucleases have the same tertiary structure. [1]

- **Same primary structure / amino acid sequence, thus same folding / same types of bonds formed between R groups, resulting in same 3-D conformation.**

In an investigation on the importance of the tertiary structure of enzymes, Ribonuclease was treated with a mixture of mercaptoethanol and urea. After treatment, the enzyme activity was measured. The results are shown in Fig. 1.2.

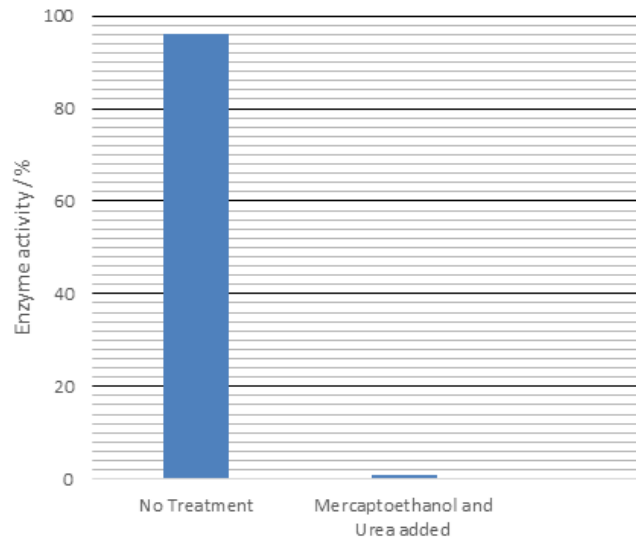


Fig. 1.2

(b) Using information from Fig. 1.1 and 1.2, explain the enzyme activity after mercaptoethanol and urea were added. [3]

- **Enzyme as activity decrease from 96% to 1%**
- **due to breaking of disulphide bonds resulting in lost 3-D conformation / denaturation**
- **and Ribonuclease can no longer bind to RNA / no RNA binding site / no active site**

Using the radioactive-labelled DNA, a cell with condensed chromosomes was carefully observed and the following measurements were taken:

A – distance from pole to pole

B – distance between a centromere and the respective poles

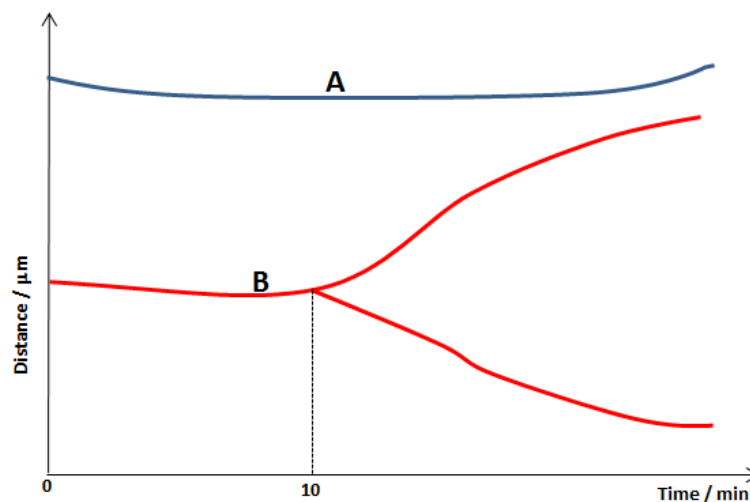


Fig. 1.3

(c) Describe the event that occurred in **B** after 10 minute. [3]

- After 10 min, the **centromere** of the chromosome **divides**
- and the **sister chromatids** are **pulled apart** by the **shortening of the centromere-to-poles spindle fibres** / microtubules.
- Hence, the **distance between the centromere and one of the poles** that it is pulled towards will **decrease** while it will **increase** with respect to the **other pole** that it is pulled away from (OWTTE)

(d) Some drugs which are used in the treatment of cancer affect the function of spindle fibres. Suggest what would happen if the drug is introduced into the cell at 0 min. [2]

- **spindle fibre not attached** to kinetochore during **metaphase** / **spindle fibre cannot be formed** during **prophase**
- resulting in **arrest of cell division** (OWTTE)

[Total: 10]

2 **Fig. 2.1** shows a molecule of transfer RNA (tRNA) attached to a corresponding enzyme.

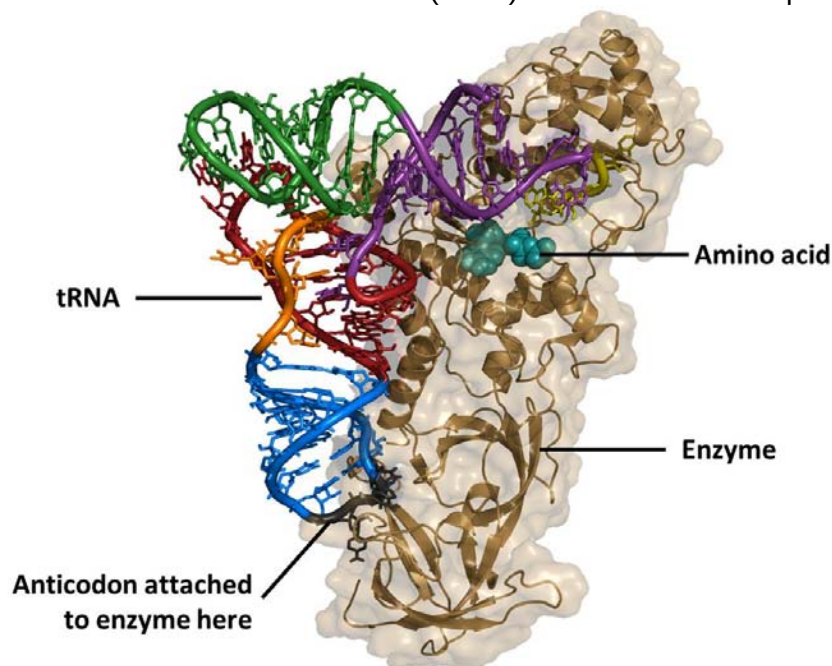


Fig. 2.1

(a) Identify the enzyme labelled in Fig. 2.1. [1]

- **Aminoacyl tRNA synthetase**

(b) Describe how the structure of the tRNA molecule and the enzyme molecule is maintained.

[2]

structure of the tRNA maintained by:

- **Hydrogen bonds between complementary base pairs / A-U, G-C [C]**

structure of the enzyme maintained by:

- **Tertiary structure - Hydrogen bonds, hydrophobic interactions, disulfide bonds and ionic bonds, between R groups of amino acids [C]**

OR

- **Secondary structure – hydrogen bonds between peptide bond regions / C=O and N-H groups of peptide bond regions**

- (c) A specific example of the enzyme in Fig. 2.1 allows the amino acid histidine to bind to its active site. A single base substitution occurred in the gene coding for this enzyme results in the enzyme recognizing tyrosine instead of histidine.

Suggest why a mutation in this gene would have extremely damaging effects on the organism. [3]

- The enzyme would **attach tyrosine to the tRNA molecule with anticodon** that is for **histidine**
- **All resulting polypeptides/proteins would have histidine replaced by tyrosine.**
- (OWTTE) **Bonds between R groups of amino acid molecules / folding / 3D structure / properties of proteins altered and become less or non-functional.**
- **essential cellular functions / metabolism cannot occur**

- (d) The flow of genetic information from DNA to polypeptide is an accurate process. Explain how this is so. [4]

- By **complementary base pairing, triplets of DNA / template is transcribed into mRNA.**
- During translation, non-overlapping triplets of **mRNA codons** are matched by **tRNAs with complementary anticodons**
- Each **tRNA carries a specific amino acid / Aminoacyl-tRNA synthetase** attach the **specific amino acid to each type of tRNA**
- This enzyme has **active site complementary to the specific amino acid to be attached and has recognition site for the correct anticodon of the tRNA.**

[Total: 10]

3 Fig. 3.1 summarises the reactions which occur in the Calvin cycle.

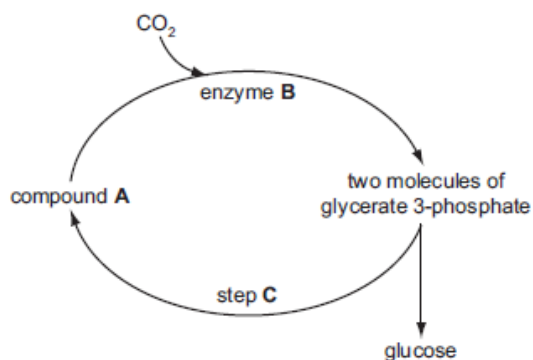


Fig. 3.1

(a) Name

(i) Compound A [1]

- ribulose biphosphate / RuBP

(ii) Enzyme B [1]

- RUBISCO / ribulose biphosphate carboxylase oxygenase

The Calvin cycle is part of the light-independent reactions of photosynthesis. These reactions continue when a plant is moved from light conditions to dark conditions, but only for a very short time.

(b) With reference to Fig. 3.1, explain why this is the case. [3]

- Initially, **ATP and reduced NADP** from light reactions are **still available**
- With time, in the absence of light, **became depleted**. Calvin cycle stops because
- ATP not available for **regeneration of RuBP**,
- ATP and reduced NADP are not available for **reduction of 1,3 bishphosphoglycerate to triose phosphate/glyceraldehyde-3-phosphate**,

Fig. 3.2 outlines the early stages of respiration

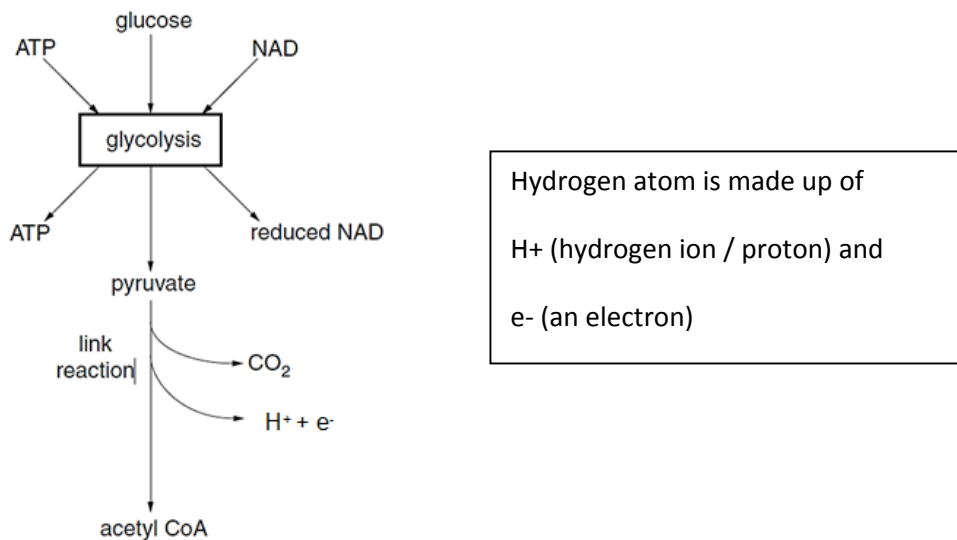


Fig. 3.2

With reference to Fig. 3.2,

(c) (i) explain why ATP is needed at the start of glycolysis. [1]

- To **phosphorylate** the glucose/fructose-6-phosphate/hexoses
- To make the glucose molecule **more active**

(ii) explain the fate of the hydrogen atoms released by pyruvate during the link reaction. [3]

- H atoms are **accepted by NAD to form reduced NAD/NADH**.
- Electrons are passed along the **electron transport chain** of progressively lower energy level to provide energy for active transport of H⁺.
- H⁺ **actively transported** into the intermembrane space via **proton pump** to generate **proton gradient** and **diffuse** via **ATP synthase** back to the matrix to generate **ATP**.
- Eventually, **H⁺ and electron combine with oxygen to form water**.

(d) State why ATP is regarded as the universal energy currency in organisms. [2]

- ATP is **found in all organisms/ all cells** ;
- **small** and **water soluble** so can be **easily transported around cell**;
- Hydrolysis of phosphate **releases energy** (30.5 kJ per mole) **for cellular processes**;
- One example - active transport / muscle contraction / Calvin cycle / protein synthesis, etc ;
- Formation of ATP from ADP + Pi is **reversible** / readily inter-convertible between ATP and ADP

[Total: 11]

4 A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 4.1.

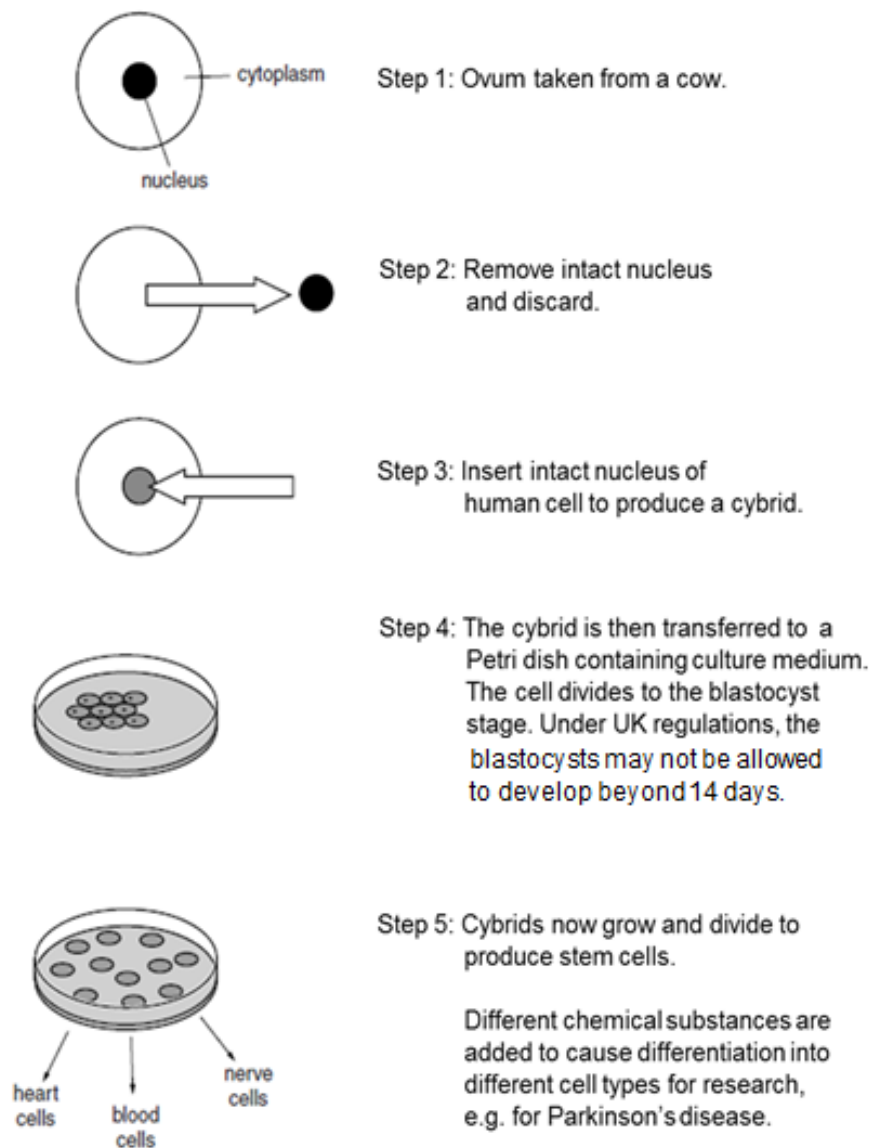


Fig. 4.1

(a) (i) Describe the unique features of the stem cells obtained in step 4. [2]

- The embryonic stem cells produced are **pluripotent**, have the **ability to differentiate into all types of cells except the extra embryonic membrane**.
- **Unspecialised/ undifferentiated**
- Can be **induced to differentiate** into specialised cells.
- **Divide / self-renew continually**
- **Telomerase activity present.**

(ii) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is in the cytoplasm. State why there is DNA in the cytoplasm of the cybrid. [1]

- The DNA in the cytoplasm is the **mitochondrial DNA** of the cow.

- (iii) The development of cybrids is an important source of stem cells to be used in medical researches. However, some people argue that it is unethical to allow the production of cybrid.

Suggest why the production of cybrids may be considered unethical. [1]

- Presence of human and animal DNA would mean a **human-animal hybrid, which will not happen in nature.**
- **Religious beliefs** may be compromised by **insertion of cells containing animal DNA** into the patient's body.
- Involves **destruction of embryonic cells** from the blastocyst which could develop into a fetus. Hence, **this constitutes killing potential lives.**
- Current use of cybrid might serve medical purpose, **but abuse of such technology is possible in future.**

- (b) A recombinant plasmid was produced by inserting a gene into a 19kb plasmid at *EcoRI* restriction site. The 19kb plasmid and the recombinant plasmid was then cleaved separately using a different restriction enzyme *HpaI*. The resultant fragments were then separated using gel electrophoresis. The results are shown below:

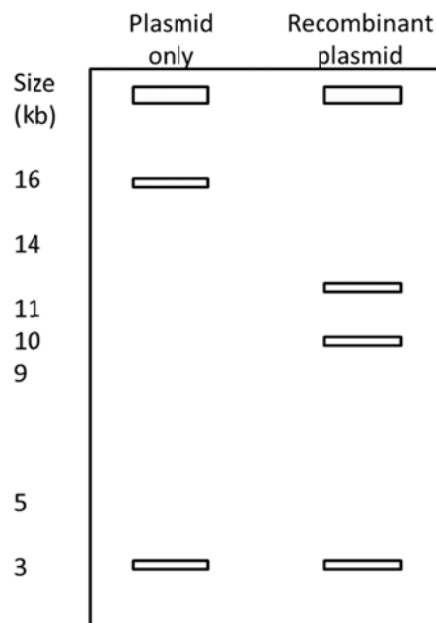
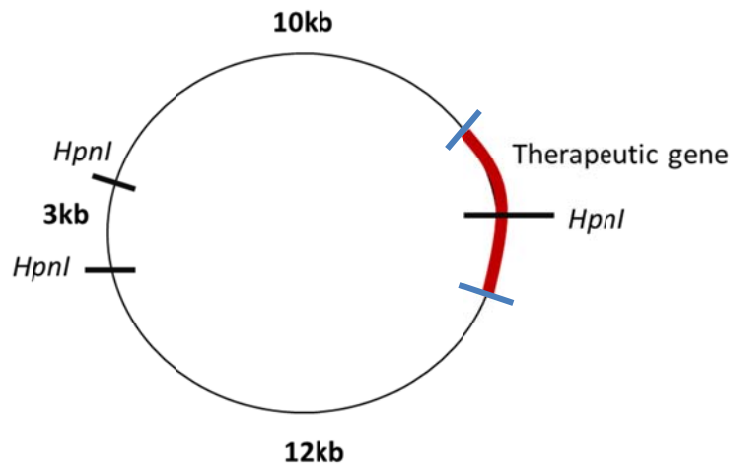


Fig. 4.2

- (i) Based on the information provided, draw and label the recombinant plasmid produced by the scientist. On your diagram, clearly show the positions of the restriction sites. [2]



- 3 *HpnI* restriction sites (labelled). One must cut at the therapeutic gene.
- Therapeutic gene flanked by *EcoRI*.

(ii) Outline how the band pattern in Fig. 4.2 was obtained. [3]

- Plasmid and recombinant plasmid **cleaved by *HpnI*** restriction enzyme
- DNA fragments from each same **separated by gel electrophoresis** where negatively-charged DNA **move towards the anode** (positive pole)
- **Smaller fragments travel faster/further down the gel** (vice versa).
- DNA fragments are then **stained with ethidium bromide/DNA binding dye** and **visualise under UV light**.

[Total: 9]

QUESTION 5

(a) Describe the similarities and differences between homologous chromosomes. [6]

Similarities

1. Homologous chromosomes have **identical gene loci**, which determine the same characteristics.
2. Homologous chromosomes have the **same number of genes** arranged along the chromosomes, and each is approximately the same length.
3. The **centromeres** of the homologous chromosomes lie in the **same position** as their sequence of genes on the chromosomes are arranged similarly.
4. Same chromatid length

Differences

5. Each homologous chromosome contains **genes from different parents**, one set of genes from father and another from mother.
6. Homologous chromosomes may have **different alleles**, due to inheritance of one chromosome from each parent.

7. **Allele** on one homologous chromosome may be **dominant / recessive** to the allele on the other homologous chromosome.

(b) Outline the process of the photolysis of water and describe what happens to the products of photolysis. [6]

1. Photosystem II absorbs light/photon
2. Enzyme in PII breaks down water to form **oxygen, proton/hydrogen and electrons**
3. Oxygen is used by cells for (aerobic) respiration
4. Electrons are used to replace electrons loss from PS II
5. protons and electrons from from PI are used to **reduce NADP to form reduced NADP**.
6. **Reduced NADP** is used in light independent stage / Calvin cycle which reduced **Glycerate 3 Phosphate to Triose Phosphate**
7. Electrons released by photolysis of water is used in ETC to **release energy for photophosphorylation** and to produce **ATP via ATP synthase**
8. H⁺ ions from photolysis of water **establish proton gradient** for synthesis of ATP via ATP synthase

(c) Discuss the implications of Genetically Modified Organisms. [8]

1. Vectors that are used for transforming plant cells (e.g. tomatoes) contain **genes for antibiotic resistance**. The gene may **pass** from the genetically modified tomato to *E. coli* and other **bacteria** in the gut when ingested, making them **resistant to the antibiotics**.
2. **Allergies** may arise from the genetically modified plants carrying **genes encoding** antigenic proteins which could be **allergens** e.g. peanuts and brazil nuts gene in soy bean products.
3. **Genetically modified crop plants** may be more hardy and **become agricultural weeds** that invade natural habitats, e.g. tomato plants that contain pest-resistance gene.
4. The **introduced transgene(s) to wild species** may result in **more invasive hybrid offspring**.
5. **Risk of toxins** produced by GM crop **affecting non-target organisms**, **upsetting the natural balance of the ecological system**.
6. **Lack of labeling** of **GM food** products thus they are **unable to make an informed choice**.
7. **GM food products** may be **unprofitable** if consumers avoid GM foods due to **fear / lack of education** on GM food.
8. **Plants** that have been **modified to carry animal genes** may be a **concern for the vegetarians / certain religious groups**.
9. The genetically modified **animals may be stressed** with the increased production of milk, meat, eggs and other products. In such cases, the **well-being of the animal was compromised for commercial gains**.
10. **Farmers** right now have to pay **royalties** to sow the crops developed by agricultural companies, thus **erode farmers' income / may also raise prices of crops**.
11. The techniques will used in creating GMO may be applied in **cloning humans** which **violate ethical code of conduct**.

QUESTION 6

(a) Describe the similarities and differences between a secretory vesicle and a lysosome. [6]

Similarities - Both

1. Bound by single membrane
2. bud off / derived from the Golgi apparatus
3. may contain digestive/hydrolytic enzymes

Differences

feature		
contains	4. enzymes / hormones / glycoprotein	hydrolytic enzymes only
roles	5. Transport of substances / enzymes / hormones //glycoprotein to the outside of the cell 6. by exocytosis 7. Replenishes cell surface membrane	Intracellular digestion autophagy, autolysis, digestion of engulfed food does not replenishes cell surface membrane

(b) Outline how the frequency of sickle cell anaemia allele is maintained in malaria affected area. [8]

1. Diploidy condition
2. Heterozygote advantage
3. sickle cell trait, thus, malaria parasite / Plasmodium less able to infect sickle cell
4. Homozygous recessive, sickle anaemia
5. red blood cells of sickle cell anaemia individuals cannot carry oxygen, death resulted
6. Homozygous dominant, normal RBC but susceptible to malaria parasite
7. Heterozygous / sickle cell traits individuals are at **selective advantage** in malaria affected areas / more resistant to malaria
8. Heterozygous individuals **survive to reproduce** and **pass on the sickle cell allele** to **viable and fertile offspring**, thus **maintain/increase the allele frequency of sickle cell allele**
9. **Frequency of sickle cell allele is higher in malaria affected areas**

(c) Discuss the ethical concerns for the use of human genome project. [6]

1. **Fairness in the use of genetic information** by insurers, etc which may create a situation in which **less healthy individuals are marginalised**.
2. There will be issues pertaining to the **confidentiality and privacy of genetic information** as the right of ownership of the genetic information has yet to be determined.
3. Genetic information may be used in controversial decisions like reproduction/ gene therapy /termination of pregnancies. Hence resulting in the ethical issue of one generation **determining the next generation's genotype before birth without their consent**.

4. Selective breeding may lead to an acute bioethical problem; inadequate education may lead to the decline in appreciation of the dignity of life resulting in widespread abortion.
5. **Clinical issues** including the doctors, other health service providers, patients and the public must be **educated** to make **informed choices**, and **be aware of scientific capabilities** and **limitations**.
6. **Conceptual and philosophical implications** regarding human responsibility, free will vs genetic determinism, and concepts of health and disease.
 - a. E.g. Difficult to determine if a person's behavior is genetically determined or can it be controlled by the that individual.
7. **Commercialization/Patenting of products** may **limit their accessibility and development of useful products**.

THE END